

Table 1: Complete performance comparison across algorithms and step budgets.

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit	
SafePO	1c3s5z	7.64 ± 0.16	9.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	1c3s5z	17.54 ± 0.94	7.03 ± 1.70	0.36 ± 0.27	500k		0
MAPPO-Lag	1c3s5z	19.82 ± 0.15	2.22 ± 0.64	0.96 ± 0.04	1M		0
MAPPO-Lag	1c3s5z	20.00 ± 0.00	0.44 ± 0.11	1.00 ± 0.00	5M		0
Safe Dreamers	1c3s5z	19.29 ± 0.14	4.23 ± 0.27	0.82 ± 0.02	100k		0
SafePO	1c3s5z	7.79 ± 0.05	9.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	1c3s5z	17.24 ± 1.16	7.19 ± 1.71	0.32 ± 0.30	500k		4
MAPPO-Lag	1c3s5z	19.82 ± 0.16	2.08 ± 0.66	0.96 ± 0.04	1M		4
MAPPO-Lag	1c3s5z	20.00 ± 0.00	0.37 ± 0.14	1.00 ± 0.00	5M		4
Safe Dreamers	1c3s5z	19.15 ± 0.53	4.83 ± 0.94	0.79 ± 0.12	100k		4
SafePO	2m_vs_1z	2.71 ± 0.18	2.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	2m_vs_1z	19.97 ± 0.02	0.00 ± 0.00	1.00 ± 0.00	500k		0
MAPPO-Lag	2m_vs_1z	19.95 ± 0.01	0.01 ± 0.00	1.00 ± 0.00	1M		0
MAPPO-Lag	2m_vs_1z	20.00 ± 0.01	0.00 ± 0.00	1.00 ± 0.00	5M		0
Safe Dreamers	2m_vs_1z	19.30 ± 0.49	0.14 ± 0.08	0.95 ± 0.04	100k		0
SafePO	2m_vs_1z	3.13 ± 0.01	2.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	2m_vs_1z	19.99 ± 0.01	0.00 ± 0.00	1.00 ± 0.00	500k		1
MAPPO-Lag	2m_vs_1z	19.96 ± 0.00	0.01 ± 0.00	1.00 ± 0.00	1M		1
MAPPO-Lag	2m_vs_1z	20.00 ± 0.01	0.00 ± 0.00	1.00 ± 0.00	5M		1
Safe Dreamers	2m_vs_1z	19.34 ± 0.38	0.14 ± 0.09	0.95 ± 0.03	100k		1
SafePO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	2s3z	14.54 ± 1.35	4.42 ± 0.60	0.20 ± 0.21	500k		0
MAPPO-Lag	2s3z	18.15 ± 1.96	2.88 ± 1.86	0.59 ± 0.51	1M		0
MAPPO-Lag	2s3z	19.95 ± 0.05	0.28 ± 0.15	0.99 ± 0.01	5M		0
Safe Dreamers	2s3z	18.44 ± 0.25	2.28 ± 0.38	0.77 ± 0.04	100k		0
SafePO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	2s3z	13.66 ± 1.30	4.68 ± 0.48	0.11 ± 0.17	500k		4
MAPPO-Lag	2s3z	18.57 ± 1.52	2.26 ± 1.10	0.74 ± 0.30	1M		4
MAPPO-Lag	2s3z	19.96 ± 0.02	0.18 ± 0.06	0.99 ± 0.01	5M		4
Safe Dreamers	2s3z	17.37 ± 1.38	2.98 ± 0.64	0.63 ± 0.15	100k		4
SafePO	2s_vs_1sc	2.44 ± 0.10	2.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	2s_vs_1sc	19.69 ± 0.47	0.62 ± 0.25	0.93 ± 0.07	500k		0
MAPPO-Lag	2s_vs_1sc	20.17 ± 0.01	0.21 ± 0.01	1.00 ± 0.00	1M		0
MAPPO-Lag	2s_vs_1sc	20.23 ± 0.01	0.11 ± 0.07	1.00 ± 0.00	5M		0
Safe Dreamers	2s_vs_1sc	19.95 ± 0.21	0.23 ± 0.11	0.96 ± 0.03	100k		0
SafePO	2s_vs_1sc	4.21 ± 0.21	2.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	2s_vs_1sc	18.79 ± 1.35	0.94 ± 0.23	0.83 ± 0.16	500k		1
MAPPO-Lag	2s_vs_1sc	20.17 ± 0.01	0.18 ± 0.04	1.00 ± 0.00	1M		1
MAPPO-Lag	2s_vs_1sc	20.23 ± 0.00	0.10 ± 0.03	1.00 ± 0.00	5M		1
Safe Dreamers	2s_vs_1sc	19.57 ± 0.49	0.31 ± 0.10	0.92 ± 0.06	100k		1
SafePO	3m	2.06 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3m	18.76 ± 0.90	0.92 ± 0.58	0.90 ± 0.07	500k		0
MAPPO-Lag	3m	19.90 ± 0.05	0.16 ± 0.13	0.99 ± 0.00	1M		0
MAPPO-Lag	3m	19.99 ± 0.01	0.02 ± 0.01	1.00 ± 0.00	5M		0
Safe Dreamers	3m	19.57 ± 0.25	0.35 ± 0.14	0.97 ± 0.02	100k		0
SafePO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3m	19.83 ± 0.16	0.19 ± 0.14	0.99 ± 0.01	500k		1
MAPPO-Lag	3m	19.93 ± 0.02	0.05 ± 0.01	1.00 ± 0.00	1M		1
MAPPO-Lag	3m	20.00 ± 0.00	0.02 ± 0.00	1.00 ± 0.00	5M		1

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit	
Safe Dreamers	3m	19.46 ± 0.42	0.60 ± 0.17	0.96 ± 0.03	100k		1
SafePO	3s5z_vs_3s6z	3.47 ± 0.02	8.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s5z_vs_3s6z	9.27 ± 0.59	8.00 ± 0.00	0.00 ± 0.00	500k		0
MAPPO-Lag	3s5z_vs_3s6z	10.46 ± 0.56	8.07 ± 0.01	0.00 ± 0.00	1M		0
MAPPO-Lag	3s5z_vs_3s6z	13.42 ± 0.66	8.00 ± 0.00	0.00 ± 0.00	5M		0
Safe Dreamers	3s5z_vs_3s6z	9.60 ± 0.84	8.00 ± 0.00	0.00 ± 0.00	100k		0
SafePO	3s5z_vs_3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	3s5z_vs_3s6z	9.53 ± 0.26	8.00 ± 0.00	0.00 ± 0.00	500k		4
MAPPO-Lag	3s5z_vs_3s6z	10.57 ± 0.05	8.08 ± 0.00	0.00 ± 0.00	1M		4
MAPPO-Lag	3s5z_vs_3s6z	13.10 ± 1.06	8.00 ± 0.00	0.00 ± 0.00	5M		4
Safe Dreamers	3s5z_vs_3s6z	13.28 ± 1.10	7.54 ± 0.40	0.11 ± 0.06	100k		4
SafePO	3s_vs_3z	6.04 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s_vs_3z	10.44 ± 0.18	3.00 ± 0.00	0.00 ± 0.00	500k		0
MAPPO-Lag	3s_vs_3z	12.03 ± 0.39	3.02 ± 0.00	0.00 ± 0.00	1M		0
MAPPO-Lag	3s_vs_3z	20.01 ± 0.02	0.01 ± 0.01	1.00 ± 0.00	5M		0
Safe Dreamers	3s_vs_3z	19.83 ± 0.20	0.95 ± 0.18	0.96 ± 0.02	100k		0
SafePO	3s_vs_3z	6.28 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3s_vs_3z	10.66 ± 0.31	3.00 ± 0.00	0.00 ± 0.00	500k		1
MAPPO-Lag	3s_vs_3z	12.01 ± 0.32	3.01 ± 0.01	0.00 ± 0.00	1M		1
MAPPO-Lag	3s_vs_3z	20.02 ± 0.04	0.01 ± 0.01	1.00 ± 0.00	5M		1
Safe Dreamers	3s_vs_3z	19.93 ± 0.20	0.79 ± 0.48	0.96 ± 0.04	100k		1
SafePO	3s_vs_4z	4.39 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s_vs_4z	6.83 ± 0.12	3.00 ± 0.00	0.00 ± 0.00	500k		0
MAPPO-Lag	3s_vs_4z	7.49 ± 0.04	3.01 ± 0.01	0.00 ± 0.00	1M		0
MAPPO-Lag	3s_vs_4z	13.07 ± 0.86	3.00 ± 0.00	0.00 ± 0.00	5M		0
Safe Dreamers	3s_vs_4z	18.19 ± 1.57	2.64 ± 0.34	0.23 ± 0.16	100k		0
SafePO	3s_vs_4z	4.55 ± 0.01	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3s_vs_4z	6.87 ± 0.10	3.00 ± 0.00	0.00 ± 0.00	500k		1
MAPPO-Lag	3s_vs_4z	7.73 ± 0.17	3.02 ± 0.01	0.00 ± 0.00	1M		1
MAPPO-Lag	3s_vs_4z	12.52 ± 1.70	3.00 ± 0.00	0.00 ± 0.00	5M		1
Safe Dreamers	3s_vs_4z	16.96 ± 0.74	2.47 ± 0.35	0.31 ± 0.11	100k		1
SafePO	3s_vs_5z	3.44 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	3s_vs_5z	5.24 ± 0.59	3.00 ± 0.00	0.00 ± 0.00	500k		0
MAPPO-Lag	3s_vs_5z	5.97 ± 0.65	3.02 ± 0.01	0.00 ± 0.00	1M		0
MAPPO-Lag	3s_vs_5z	10.08 ± 0.56	3.01 ± 0.01	0.00 ± 0.00	5M		0
Safe Dreamers	3s_vs_5z	16.88 ± 1.58	2.67 ± 0.40	0.15 ± 0.19	100k		0
SafePO	3s_vs_5z	3.52 ± 0.02	3.00 ± 0.00	0.00 ± 0.00	5M		1
MAPPO-Lag	3s_vs_5z	5.51 ± 0.36	3.00 ± 0.00	0.00 ± 0.00	500k		1
MAPPO-Lag	3s_vs_5z	6.61 ± 0.14	3.02 ± 0.02	0.00 ± 0.00	1M		1
MAPPO-Lag	3s_vs_5z	9.33 ± 0.18	3.00 ± 0.00	0.00 ± 0.00	5M		1
Safe Dreamers	3s_vs_5z	17.44 ± 1.49	2.67 ± 0.43	0.17 ± 0.22	100k		1
SafePO	8m	1.43 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	5M		0
MAPPO-Lag	8m	9.09 ± 0.03	8.00 ± 0.00	0.00 ± 0.00	500k		0
MAPPO-Lag	8m	16.56 ± 0.83	5.67 ± 0.46	0.60 ± 0.08	1M		0
MAPPO-Lag	8m	19.96 ± 0.02	0.90 ± 0.17	1.00 ± 0.00	5M		0
Safe Dreamers	8m	18.64 ± 2.02	2.81 ± 1.91	0.86 ± 0.21	100k		0
SafePO	8m	1.63 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	5M		4
MAPPO-Lag	8m	9.04 ± 0.43	8.00 ± 0.00	0.00 ± 0.00	500k		4
MAPPO-Lag	8m	15.86 ± 2.19	5.96 ± 1.20	0.53 ± 0.25	1M		4
MAPPO-Lag	8m	19.95 ± 0.03	1.10 ± 0.36	0.99 ± 0.00	5M		4

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Table 1 – continued

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
Safe Dreamers	8m	17.26 $\pm$ 1.04	5.03 $\pm$ 0.42	0.69 $\pm$ 0.10	100k	4
SafePO	MMM	1.87 $\pm$ 0.01	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
MAPPO-Lag	MMM	6.44 $\pm$ 0.49	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	0
MAPPO-Lag	MMM	10.58 $\pm$ 0.23	10.11 $\pm$ 0.01	0.00 $\pm$ 0.00	1M	0
MAPPO-Lag	MMM	19.89 $\pm$ 0.14	0.97 $\pm$ 0.61	0.99 $\pm$ 0.02	5M	0
Safe Dreamers	MMM	17.05 $\pm$ 2.36	6.77 $\pm$ 1.79	0.59 $\pm$ 0.25	100k	0
SafePO	MMM	1.89 $\pm$ 0.04	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
MAPPO-Lag	MMM	6.29 $\pm$ 1.28	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	500k	4
MAPPO-Lag	MMM	10.04 $\pm$ 0.92	10.11 $\pm$ 0.02	0.00 $\pm$ 0.00	1M	4
MAPPO-Lag	MMM	19.61 $\pm$ 0.47	1.41 $\pm$ 1.07	0.97 $\pm$ 0.03	5M	4
Safe Dreamers	MMM	17.02 $\pm$ 1.32	7.88 $\pm$ 0.16	0.55 $\pm$ 0.07	100k	4
SafePO	bane_vs_bane	19.26 $\pm$ 0.01	22.55 $\pm$ 0.02	0.79 $\pm$ 0.01	5M	0
MAPPO-Lag	bane_vs_bane	19.85 $\pm$ 0.03	14.41 $\pm$ 0.55	0.95 $\pm$ 0.00	500k	0
MAPPO-Lag	bane_vs_bane	19.89 $\pm$ 0.01	10.28 $\pm$ 0.40	0.96 $\pm$ 0.00	1M	0
MAPPO-Lag	bane_vs_bane	20.00 $\pm$ 0.00	5.17 $\pm$ 0.19	1.00 $\pm$ 0.00	5M	0
Safe Dreamers	bane_vs_bane	17.75 $\pm$ 2.82	13.91 $\pm$ 7.11	0.62 $\pm$ 0.39	100k	0
SafePO	bane_vs_bane	19.37 $\pm$ 0.09	22.40 $\pm$ 0.06	0.80 $\pm$ 0.02	5M	4
MAPPO-Lag	bane_vs_bane	19.91 $\pm$ 0.05	13.65 $\pm$ 0.88	0.96 $\pm$ 0.01	500k	4
MAPPO-Lag	bane_vs_bane	19.92 $\pm$ 0.03	9.13 $\pm$ 1.18	0.98 $\pm$ 0.01	1M	4
MAPPO-Lag	bane_vs_bane	20.00 $\pm$ 0.00	5.22 $\pm$ 0.17	1.00 $\pm$ 0.00	5M	4
Safe Dreamers	bane_vs_bane	19.91 $\pm$ 0.11	6.37 $\pm$ 1.20	0.98 $\pm$ 0.02	100k	4